

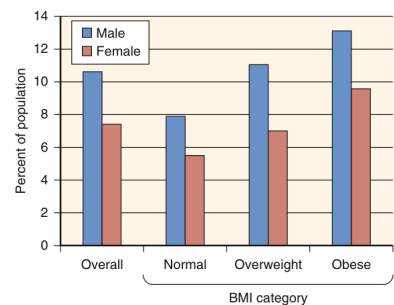


Fig. 40.1 Prevalence of urolithiasis plotted as a function of gender and body mass index (BMI). (Modified from Scales CD Jr, Smith AC, Hanley JM, Saigal CS. Urologic diseases in America. Prevalence of kidney stones in the United States. *Eur Urol.* 2012;62[1]:160–165.)

Fig_40_1

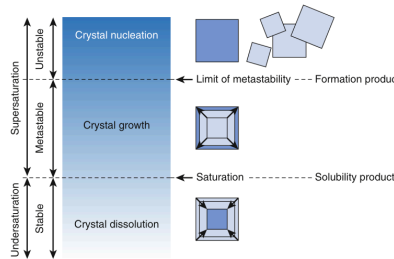


Fig. 40.4 Physicochemical parameters used in the assessment of kidney stone risk. These are shown in relation to the three states of crystal—dissolution, growth, and nucleation.

Fig_40_4

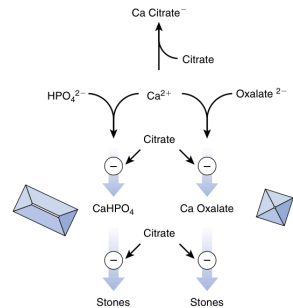


Fig. 40.7 Protective effects of citrate against urolithiasis. Citrate binds calcium with high affinity to form soluble calcium citrate complexes and lowers the ionized calcium and calcium activity. Citrate also directly inhibits calcium oxalate and calcium phosphate crystallization and aggregation.

Fig_40_7

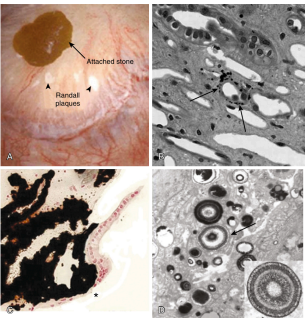


Fig. 40.2 Histopathology of Randall's plaques in calcium oxalate stone formers. (A) View during endoscopic surgery. One plaque (not visible) has an overlying attached calcium oxalate stone (arrow). Two Randall plaques without stones are below (arrowheads). (B) Light microscopic image of the basal part of the initial crystal deposits in the basement membrane of the thin loops of Henle (arrows). (C) Calcium phosphate (stained black) showing a more advanced lesion filling the interstitial space and extending (↑), eventually protruding into the urinary space and corresponding to the cream-colored plaques (arrow) assuming the morphology of multilaminated spherules of alternating layers of electron-lucent hyaline crystal and electron-dense matrix (inset). (D) The individual deposits (arrow) assume the morphology of multilaminated spherules of alternating layers of electron-lucent hyaline crystal and electron-dense matrix (inset). (E) From Evan AP, Lingeman JE, Coe FL, et al. Randall's plaque of patients with nephrolithiasis begins in basement membranes of thin loops of Henle. *J Clin Invest.* 2003;111[5]:607–616.)

Fig_40_2

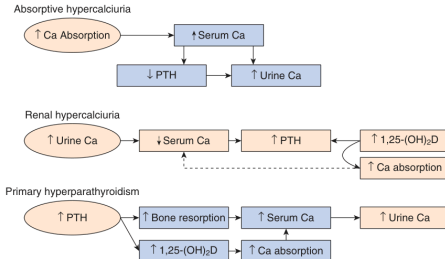


Fig. 40.5 Pathophysiologic mechanisms of hypercalciuria. PTH, Parathyroid hormone. (From Pak CY. Etiology and treatment of urolithiasis. *Am J Kidney Dis.* 1991;18[6]:624–637.)

Fig_40_5

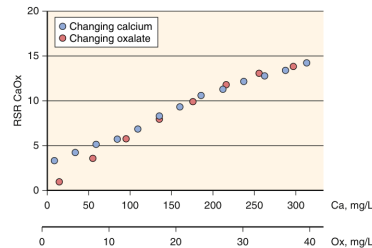


Fig. 40.8 Relationship between calcium oxalate relative supersaturation (RSR) and calcium or oxalate concentration. Calcium or oxalate was varied individually; the other was kept constant. (Modified from Pak CY, Adams-Huet B, Poindexter JR, et al. Rapid communication: relative effect of urinary calcium and oxalate on saturation of calcium oxalate. *Kidney Int.* 2004;66[5]:2032–2037.)

Fig_40_8

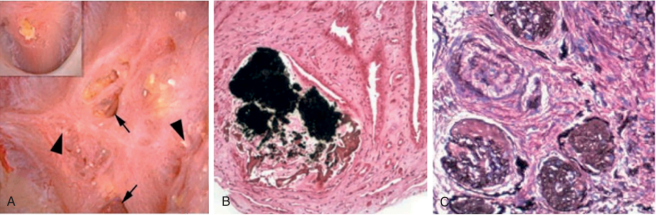


Fig. 40.3 Histopathology of calcium phosphate stone formers. (A) View during endoscopic surgery. Depressions near the papillary tips (arrows) are unique to calcium phosphate stone formers, which coexist with Randall plaques (arrow heads). The papillae show yellow crystalline deposits coming out of the ducts of Bellini (inset). (B) Micrograph of luminal deposits in an inner medullary collecting duct. The crystal deposits greatly expanded the lumen, and cell injury and necrosis were found. Interstitial inflammation and fibrosis surround the intraluminal crystal deposition. (C) Renal biopsy from a patient with calcium phosphate stones showing advanced glomerulosclerosis, tubular atrophy, and interstitial fibrosis. This is rarely seen in calcium oxalate stone formers. (From Evan AP, Lingeman JE, Coe FL, et al. Crystal-associated nephropathy in patients with brushite nephrolithiasis. *Kidney Int.* 2005;67[2]:576–591.)

Fig_40_3

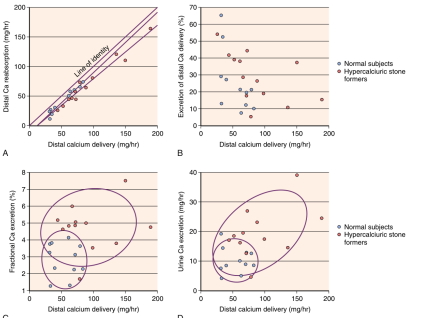


Fig. 40.6 Relationship among distal calcium reabsorption, renal calcium excretion, and distal calcium delivery. (A) Distal calcium reabsorption increases linearly with calcium delivery in normal subjects and hypercalciuric stone formers. Both lines are positioned slightly below the line of identity. This suggests normal distal calcium handling but higher distal calcium delivery. (B) The actual fraction (%) of distally delivered calcium that was excreted fell with increasing distal delivery, and stone formers lie above normal at comparable distal delivery. (C) Fractional and total (D) calcium excretion were high in many stone formers versus normal subjects at comparable deliveries. (From Vignati BA, Coe FL, Evan AP, et al. Evidence for increased postprandial distal calcium delivery in hypercalciuric stone-forming patients. *Am J Physiol Renal Physiol.* 2008;296[1]:F109–F124.)

Fig_40_6

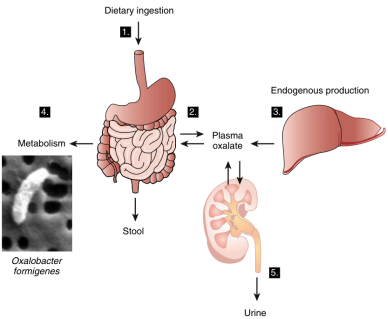


Fig. 40.9 Pathophysiologic mechanisms of hyperoxaluria. Oxalate balance is determined by ingestion and endogenous production versus intestinal and urinary excretion. Intestinal handling can be bidirectional, and luminal degradation is facilitated by microbial degradation. Urinary excretion is the net result of filtration minus tubular reabsorption. Hyperoxaluria can be caused by the following: 1. increased dietary ingestion; 2. increased gut absorption or decreased secretion; 3. increased endogenous hepatic production; 4. decreased intestinal bacterial metabolism; or 5. renal hyperexcretion.

Fig_40_9

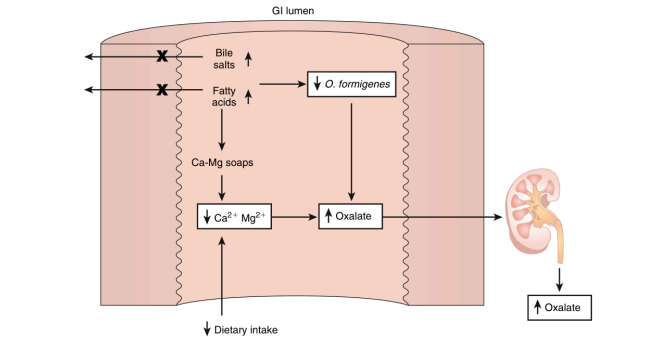


Fig. 40.10 Pathophysiologic mechanisms of hyperoxaluria in inflammatory bowel disease or following bariatric surgery. Fatty acids and bile salts precipitate luminal calcium, which can be compounded by low dietary intake. Low luminal calcium frees up and increases unbound oxalate. Lack of *Oxalobacter formigenes* because of excess bile salts also contributes to high luminal oxalate levels. *GI*, Gastrointestinal.

Fig_40_10

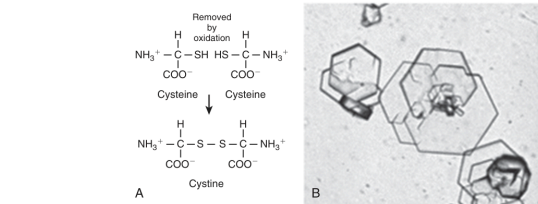


Fig. 40.13 (A) Cystine is a dimer of cysteine formed by disulfide bonding under oxidizing conditions. (B) Appearance of cystine crystals in the urine of a patient with cystinuria.

Fig_40_13

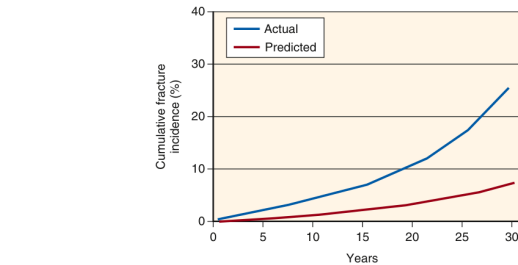


Fig. 40.16 Cumulative incidence of vertebral fractures in stone formers; data from Rochester, Minnesota, residents following an initial episode of symptomatic nephrolithiasis. The elevated fracture risk was vertebral and was present in both sexes. (Modified from Melton LJ 3rd, Crowson CS, Khosla S, et al. Fracture risk among patients with urolithiasis: a population-based cohort study. *Kidney Int.* 1998;53[2]:459–464.)

Fig_40_16

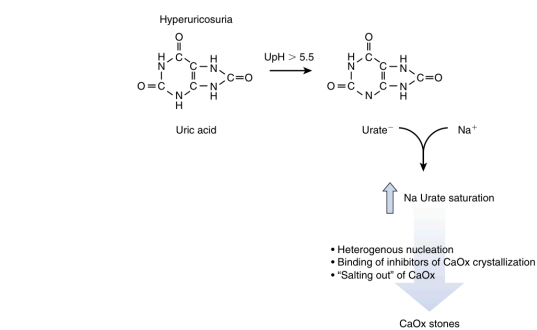


Fig. 40.11 Physicochemical scheme for urate-induced CaOx stones. Hyperuricosuria in the absence of acidic urine renders high free urate activity levels in the urine. Sodium activity is perpetually higher than urate by two orders of magnitude. The higher sodium urate promotes calcium oxalate crystallization by three nonexclusive mechanisms. *UpH*, Urinary pH.

Fig_40_11

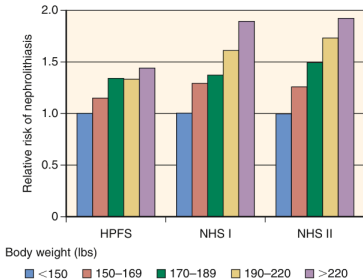


Fig. 40.14 Obesity and the risk of nephrolithiasis. The data were pooled from three databases. *HPFS*, Health Professionals Follow-Up Study; *NHS I*, Nurses' Health Study I; *NHS II*, Nurses' Health Study II. (Modified from Sakhaee K, Maalouf NM, Sirtori B. Clinical review. Kidney stones 2012: pathogenesis, diagnosis, and management. *J Clin Endocrinol Metab.* 2012;97[6]:1847–1860.)

Fig_40_14

Table 40.1 Causes and Mechanisms for Uric Acid Nephrolithiasis			
Causative Factors	Low Urine Volume	Low Urinary pH	Hyperuricosuria
Acquired Conditions			
Diarrhea	+	+	+
Myeloproliferative		+	+
High animal protein		+	+
Uricosuric drugs			+
Primary gout		+	+
Metabolic syndrome		+	
Monogenic Disease			
Enzyme disorders of purine metabolism			+
Pathogenic variants in uric acid transporters			+

Table_40_1

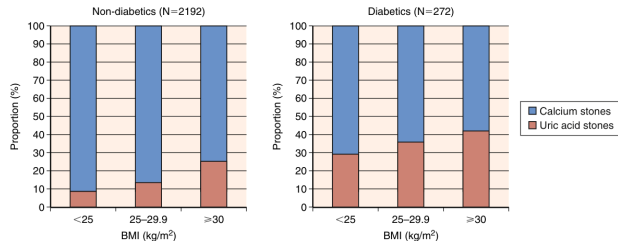


Fig. 40.12 Distribution of stone type with respect to body mass index (BMI) and diabetes status. (Based on data from Daudon M, Traxer O, Conort P, et al. Type 2 diabetes increases the risk of uric acid stones. *J Am Soc Nephrol.* 2006;17[7]:2026–2033.)

Fig_40_12

Incidence (no. at risk)

Control	0 (10,860)	0.8 (6,689)	2 (3,184)	4.2 (1,010)
Stone formers	0 (4,564)	1.3 (2,686)	3 (1,276)	5.2 (404)

Fig. 40.15 Increased risk for myocardial infarction in stone formers. Data collected from Olmsted County, Minnesota, residents. (Modified from Rule AD, Roger VL, Melton LJ 3rd, et al. Kidney stones associate with increased risk for myocardial infarction. *J Am Soc Nephrol.* 2010;21[10]:1641–1644.)

Fig_40_15

Table 40.2 Mendelian Forms of Nephrolithiasis				
Gene	Protein	Disease	Mode of Inheritance	Phenotype
AGXT	Alanine-glucate aminotransferase	Primary hyperoxaluria type 1	Autosomal recessive	Hyperoxaluria (>80 mg or >1000 μmol/24 h), calcium oxalate monohydrate stones
APRT	Adenosine phosphoribosyltransferase	Dihydropyrimidinuria	Autosomal recessive	2,8-dihydroxyadenine stones, CKD, rare need for kidney transplant
ATP6V1B1	ATPase, H ⁺ transporting, vacuolar, V1 subunit B1	Dial renal tubular acidosis x	Autosomal recessive, incomplete dRTA in some heterozygotes	Metabolic acidosis, alkaline urine pH, CaP stones, nephrocalcinosis
ATP6V0A4	ATPase, H ⁺ transporting, vacuolar, V0 subunit a4	Dial renal tubular acidosis x	Autosomal recessive	Metabolic acidosis, alkaline urine pH, CaP stones, nephrocalcinosis
CA2	Carbonic anhydrase II	Proximal and distal renal tubular acidosis	Autosomal recessive	Metabolic acidosis, alkaline urine pH, nephrocalcinosis, osteopenia
CASR	Calcium-sensing receptor (activating mutations)	Autosomal-dominant hypocalcemia, Bartter syndrome type 5	Autosomal dominant	Hypocalcemia, hypokalemia, normal PTH, calcium stones
CASR	Calcium-sensing receptor (inactivating mutations)	Familial hypocalcemic hyperkalemia	Autosomal dominant	Hypocalcemia, hypokalemia, normal to high PTH, calcium stones
CLCN6	Chloride channel, voltage-sensitive 6	Dent disease type 1	X-chromosomal recessive	Low-molecular weight proteinuria, CaP stones, nephrocalcinosis
CLCNKB	Chloride channel, voltage-sensitive Kb	Bartter syndrome type 3	Autosomal recessive	Hypokalemia, metabolic alkalosis, hypercalciuria, CaP stones
CLDN2	Claudin-2	Calcium nephrolithiasis	X-linked	Hypercalciuria and nephrolithiasis
CLDN16	Claudin-16	Familial hypomagnesemia with hypocalcemia and nephrocalcinosis	Autosomal recessive	Renal calcium and magnesium wasting
CLDN19	Claudin-19	Familial hypomagnesemia with hypocalcemia and nephrocalcinosis	Autosomal recessive	Renal calcium and magnesium wasting, ocular abnormalities
CYP26A1	Cytochrome P450 26A1/hydroxysteroid 24-hydroxylase	Interventricular hypocalcemia	Autosomal recessive	Hypercalciuria, hypocalcemia, elevated 1,25 (VLD), suppressed PTH

Table_40_2_part1

Table 40.2 Mendelian Forms of Nephroblastoma (Continued)						
Gene	Protein	Disease	Mode of Inheritance	Phenotype	Treatment	
FAM63A	Fam63 protein similarity 25, member 1	Enamel navel syndrome	Autosomal recessive	Nephroblastosis, amelogenesis imperfecta	Monitor for CKD	
FOO1	Farly with sex protein 11	Dental enamel tubular ectodermal dysplasia	Autosomal recessive	Nephroblastosis, amelogenesis imperfecta, shallow urethral pH	Alkali therapy, monitor for CKD	
GRHR	Glyoxylate reductase/hydroxyphenylpyruvate dioxygenase	Primary hyperparathyroidism	Autosomal recessive	Nephroblastosis, calcium stones	Hydration and alkali therapy	
HNF4A	HNF4A (p.R107C) HNF4A nuclear factor, alpha (p.R107C)	MODY, Fanconi syndrome, nephroblastosis	Autosomal dominant	Fanconi syndrome, hypophosphatemia, hypoglycemia, hypokalemia, hypomagnesemia, hypocalcemia	Monitor for CKD, treat hyperparathyroidism	
HNF1A	4-hydroxy-2-naphtholate aldolase 1	Primary hyperparathyroidism	Autosomal recessive	Nephroblastosis, calcium stones	Hydration and alkali therapy	
HPRT1	Hypoxanthine phosphoribosyltransferase 1	Lesch-Nyhan syndrome/ Rall-Saenger syndrome	X-chromosomal recessive	Hypoxanthinemia, UA, stone, neurologic features	Xanthine oxidase inhibitors, purine restriction	
ICNT1	Potassium ion-transporting sodium channel, isoform 1	Barter syndrome type 2	Autosomal recessive	Hypokalemia, metabolic alkalosis, hypomagnesemia, low sodium	Mineral supplementation, low sodium diet	
OCRL	Inositol polyphosphate 5-phosphatase	Deaf disease type 2 (Low syndrome) X-chromosomal recessive	X-chromosomal recessive	Low sodium, low calcium, low phosphate	Monitor for CKD and stones	
PPHS1	Phosphoribosyl pyrophosphate synthetase	Phosphoribosyl pyrophosphate synthetase deficiency	X-chromosomal recessive	Uric acid hyperuricemia, gout, kidney stones, CKD	Allopurinol, xanthine oxidase inhibitors, uric acid restriction	
SLC23A5	Gut/Slc family member 25 (hectadactyl)	Rare X-chromosomal type	Autosomal recessive	Hypokalemia, hypomagnesemia, hypocalcemia	Hydration, vitamin D, xanthine oxidase inhibitors	
SLC31A1	Basal amino acid transporter system Y1	Cystinuria (type A)	Autosomal recessive	Elevated urine cystine, cystine stones	Hydration, protein and alkalinization therapy, thiazide, monitor for CKD	
SLC6A1	Solute carrier family 6, anion exchanger, member 1	Dental enamel tubular ectodermal dysplasia (necrotic)	Autosomal dominant	Metabolic acidosis, shallow urethral pH	Alkali therapy, monitor for CKD	

Continued on following page

Table_40_2_part2

[illegible]

Table 40.4

Gene	Protein	Disease	Mode of Inheritance	Phenotype	Treatment
SLC7A9	Basic amino acid transporter system y ⁺ L	Cystinuria (type B)	Autosomal-recessive, autosomal-dominant with incomplete penetrance	Elevated urine cystine, cystine stones	Hydration, protein and sodium restriction, alkalization therapy, thiol monitor for CKD
SLCBAR1	Solute carrier family 8, subfamily A member 3 regulator 1	Hypophosphatemic nephropathias, osteopetrosis 2	Autosomal-dominant	Calcium stones, low bone density	
SLC12A1	Solute carrier family 12, member 1	Bartter syndrome type 1	Autosomal-recessive	Hypokalemia, metabolic alkalosis, hypercalciuria, CaP stones	Mineral supplementation, NSAID
SLC22A12	Urat1, solute carrier family 22 (organic anion) transporters, member 12	Renal hypouricemia type 1	Autosomal-recessive type 1	Hyperuricosuria, elevated 1,25(OH) ₂ D, calcium stones (BIAK)	Hydration, alkali, xanthine oxidase inhibitors to prevent ESKAF
SLC24A4 ¹	Sodium-dependent phosphate cotransporter 4 (NPT2c)	Hypophosphatemic nephropathias, osteopetrosis-1, osteopetrosis interfolle hypercalcaemia	Autosomal-recessive; evidence suggesting incomplete dominance	Hypophosphatemia, elevated 1,25(OH) ₂ D, hypercalciuria	Phosphate supplementation, restrict excess vitamin D, bone density measurements
SLC34A3 ²	Sodium-dependent phosphate cotransporter 3a (NPT2c)	Hypophosphatemic nephropathies with rickets containing proteinuria	Autosomal-recessive; evidence suggesting incomplete dominance	Nephropathy, hypophosphatemia, elevated 1,25(OH) ₂ D, hypercalciuria, nephrocalcinosis, calcium stones	Phosphate supplementation, restrict excess vitamin D, bone density measurements
WDR72 ³	WD-repeat containing protein T2	Distal renal tubular acidosis	Autosomal-recessive	Metabolic acidosis, alkaline urine pH, nephrocalcinosis, amelogenesis imperfecta	Alkali therapy
XDR	Xanthine dehydrogenase	Xanthinuria type 1	Autosomal-recessive	Xanthinuria, stones, hypouricaemia	Purine restriction, hydration, radiolucent stones

¹Genes also found as candidate loci in genome-wide association studies on kidney stone disease.

²Familial stones and nephrocalcinosis have only been observed with a specific allele p.R167G.

³CaP: Calcium phosphatide; ESKAF: exercise-induced acute kidney injury; CKD: chronic kidney disease; MOPV: maturity onset diabetes of the young; NSAID: nonsteroidal anti-inflammatory drug; ATP7A: paraflagellar homology, UA: uric acid.

Adapted from Pakizadeh S, Singh SS, Griffith DS, Weiss JP Personalized intervention in monogenic stone formers. J Urol 2018;190(6):623–632 and Hablitzler J. Genetics of kidney stone diseases – polygenic meets monogenic. Nephrol Ther 2021;17:S58–S64.

*Genes also found as candidate loci in genome-wide association studies on kidney stone disease.
 *Fanconi syndrome and nephrocalcinosis have only been observed with a specific allele: p.R76W.
 CaP, Calcium phosphate; EIAK, exercise-induced acute kidney injury; CKD, chronic kidney disease; MODY, maturity onset diabetes of the young; NSAID, nonsteroidal anti-inflammatory drug.
 PTH, parathyroid hormone; UA, uric acid.
 Adapted from Polcastro LJ, Saggi SJ, Goldfarb DS, Weiss JP. Personalized intervention in monogenic stone formers. *J Urol*. 2018;199(3):623–632 and Halbritter J. Genetics of kidney stone disease—polymers meet monogenic. *Nephrol Ther*. 2021;17S:S88–S94.

Table_40_2_part3

Table 40.5 Urinary Supersaturation Estimation ^a		
Term	Definition	Readout
Relative supersaturation ratio (RSR)	Ion concentration product in a given solution (activity product); ion concentration product at equilibrium (solubility product)	Values <1 represent undersaturation; values >1 represent supersaturation
Relative supersaturation (RS)	Urinary activity product in urine specimen; activity product from normal subjects	Upper normal limits = 2

^aThe EQUIL2 software program is used for the assessment of urinary supersaturation in stone research, as well as in a clinical setting. The JESS program, which is now mostly used in research programs, also represents the supersaturation ratio but is defined as saturation index in that software program.

Table 40.5

Reference	Study Participants	No. of Participants	Bone Histomorphometric Profiles
Border ¹⁰⁰	Dietary hypercalcaemia Renal hypercalcaemia Hypophosphatemia	20 19 21	None Increased osteoclast and osteoblast surfaces Increased osteoclastic and eroded surface (within normal range), decreased osteoblastic apposition rate and osteoid parameters
Mauluch et al., 1977, ¹⁰⁶	Controls Absorptive hypercalcaemia	12 15	Low-normal osteoclastic bone resorption, low osteoblastic activity, decreased formation of mineralizing osteoid seams, decreased mineralization apposition rate
de Vemeijere ¹⁰⁷	Controls Idiopathic hypercalcaemia	20 32 (20 M, 10 F)	Decreased trabecular volume, decreased active osteoblastic surface, decreased active bone resorption surface
Steiniche ¹⁰⁸	Controls Idiopathic hypercalcaemia	187 33 (22 M, 11 F)	Increased bone resorption surfaces (decreased refilling of lacunae with low bone formation), decreased bone formation rate, increased mineralization lag times
Heilberg ¹⁰⁹	Controls Fasting hypercalcaemia	30 (19 M, 11 F) 6 M	Increased eroded surface, decreased osteoid surface, decreased bone formation rate with a complete lack of tetracycline double labeling
Batala ¹⁴⁰	Controls Idiopathic hypercalcaemia	No information 24 (20 M, 4 F)	Low eroded surface, low bone volume, low osteoid surface, thickness, mineral apposition rate, adjusted apposition rate and bone formation rate
Misael da Silva ¹²¹	Controls Idiopathic hypercalcaemia	18 (9 M, 9 F) 22	High eroded surface, increased osteoblastic bone surface, no change in trabecular thickness
Heller ¹⁴¹	Control Absorptive hypercalcaemia	94 30 (18 M, 3 F)	Relatively high bone resorption (osteoclast surface, bone surface—mean value within the normal limit), lower indices of bone formation (osteoblast surface/bone surface), decreased wall thickness
	Controls	9 (6 M, 3 F)	

Table_40_3

Class of Drugs	Molecule	Indication	Dosage
Calcium-Containing Stones			
Thiazide diuretics	Hydrochlorothiazide	Hypercalcaemia	50mg/day
	Indapamide	Hypercalcaemia	2.5 mg/day
Alkali supplements	Chloramide	Hypercalcaemia	25 mg/day
	Potassium citrate	Hypocitraturia	20-60 mEq/day, divided into 2-4 times
Xanthine oxidase inhibitors	Allopurinol (first choice)	Acidic UPH	100-300 mg/day
	Febuxostat	Hyperuricaemia	100-300 mg/day
Calcium supplements	Hypercalcaemia		40-80 mg/day
	Calcium carbonate	Secondary hyperparathyroidism	500-1500 mg/day, divided into 2-3 times during meals
Uric Acid Stones			
Alkali supplements	Potassium citrate	To maintain UPH between 6.0 and 7.0	20-80 mEq/day, divided into 2-4 times
Xanthine oxidase inhibitors	Allopurinol	Hyperuricaemia	100-300 mg/day
	Febuxostat	Hyperuricaemia	40-80 mg/day
Cystine Stones			
Alkali supplements	Potassium citrate	To maintain UPH >7.0	20-80 mEq/day
Thiol compounds	Tiopronin	Ureaux cystine >1000 mg/day	750-1500 mg/day
Struvite Stones			
Antibiotic therapy	Nitrofurantoin monohydrate/ trimethoprim- sulfamethoxazole	Preoperatively/ perioperatively, to reduce the risk of urosepsis	Dosages must be adapted for each patient and involved bacteria
Urease inhibitors	Acetohydroxamic acid	In addition to antibiotics, in patients with remaining stone fragments and no other surgical options	500 mg/day, divided into 2 doses
Acidifying agents	Chlorothalidone	Inappropriately high urine pH (>7.0)	1500-3000 mg/day

Table 40_6